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State	Finished
Completed on	Saturday, 10 January 2026, 11:49 PM
Time taken	6 mins 13 secs
Marks	38.00/40.00
Grade	**95.00** out of 100.00

Question 1

Correct

Mark 1.00 out of 1.00

Which of the following is a fundamental limitation of an array?

- a. Fixed size after declaration ✔
- b. Ability to store elements of different types
- c. Inefficient random access
- d. Memory allocation is always dynamic

Question 2

Correct

Mark 1.00 out of 1.00

The time complexity of inserting an element into a binary heap is typically _____, assuming a well-balanced heap structure.

- a. O(log n) ✔
- b. O(1)
- c. O(n)
- d. O(n log n)

Question 3

Correct

Mark 1.00 out of 1.00

Which of the following is a fundamental concept of recursion in trees?

- a. A function calls itself ✔
- b. Loops are used instead of function calls
- c. Recursion is only used in binary trees
- d. A tree must be fully balanced to use recursion

Question 4

Correct

Mark 1.00 out of 1.00

Which sorting algorithm repeatedly compares and swaps adjacent elements if they are in the wrong order until the entire array is sorted?

- a. Bubble Sort ✔
- b. Quick Sort
- c. Merge Sort
- d. Selection Sort

Question 5

Correct

Mark 1.00 out of 1.00

For which operation is the front of a queue used?

- a. Removing elements ✔
- b. Adding elements
- c. Sorting elements
- d. Searching elements

Question 6

Correct

Mark 1.00 out of 1.00

Which of the following hashing techniques resolves collisions by storing multiple elements in the same bucket?

- a. Separate Chaining ✔
- b. Linear Probing
- c. Quadratic Probing
- d. Double Hashing

Question 7

Correct

Mark 1.00 out of 1.00

Which of the following problems is commonly solved using recursion?

- a. Tower of Hanoi ✔
- b. Sorting using Bubble Sort
- c. Searching in an unsorted array
- d. Hash table lookup

Question 8

Correct

Mark 1.00 out of 1.00

Which of the following is a key advantage of a stack compared to a queue?

- a. Supports Last In, First Out (LIFO) access, useful for function calls and backtracking ✔
- b. Provides faster searching of elements
- c. Allows access to elements in FIFO order
- d. Requires less memory for storing elements

Question 9

Correct

Mark 1.00 out of 1.00

What is the advantage of a linked list over an array?

- a. Dynamic memory allocation ✔
- b. Faster access to elements
- c. Fixed size allocation
- d. Requires less memory

Question 10

Correct

Mark 1.00 out of 1.00

If an algorithm requires constant space, what is its space complexity?

- a. $O(1)$ ✔
- b. $O(n)$
- c. $O(n \log n)$
- d. $O(n^2)$

Question 11

Correct

Mark 1.00 out of 1.00

The adjacency list representation is preferred when the graph is _______.

- a. Sparse ✔
- b. Dense
- c. Cyclic
- d. Complete

Question 12

Correct

Mark 1.00 out of 1.00

Which algorithm can be used as an alternative to Dijkstra's algorithm for graphs with negative weights?

- a. Bellman-Ford Algorithm ✔
- b. Kruskal's Algorithm
- c. Floyd-Warshall Algorithm
- d. Prim's Algorithm

Question 13

Correct

Mark 1.00 out of 1.00

What is the maximum height difference allowed between left and right subtrees in an AVL tree?

- a. 1 ✔
- b. 0
- c. 2
- d. No limit

Question 14

Correct

Mark 1.00 out of 1.00

What is the time complexity of DFS in an adjacency list representation?

- a. O(V + E) ✔
- b. O(V)
- c. $O(V^2)$
- d. O(E log V)

Question 15

Correct

Mark 1.00 out of 1.00

What is the time complexity of searching for an element in an unsorted singly linked list?

- a. O(n) ✔
- b. O(1)
- c. O(log n)
- d. $O(n^2)$

Question 16

Incorrect

Mark 0.00 out of 1.00

A cloud service needs to sort a dataset in memory, but available RAM is very limited. Which sorting algorithm is most suitable for this scenario due to its minimal additional memory usage?

- a. Quick Sort
- b. Selection Sort ✖
- c. Merge Sort
- d. Bubble Sort

Question 17

Correct

Mark 1.00 out of 1.00

What is the main disadvantage of inserting a new node at the tail of a singly linked list without a tail pointer?

- a. It requires traversing the entire list to find the last node ✔
- b. It is impossible to insert at the tail
- c. The inserted node always points to the head
- d. The inserted node gets lost in memory

Question 18

Correct

Mark 1.00 out of 1.00

In a binary tree, pre-order traversal always visits the left subtree before the right subtree.

- a. True ✔
- b. False

Question 19

Correct

Mark 1.00 out of 1.00

______ is a divide-and-conquer algorithm that consistently performs well on large datasets but requires additional memory proportional to the size of the input.

- a. Merge Sort ✔
- b. Bubble Sort
- c. Quick Sort
- d. Selection Sort

Question 20

Correct

Mark 1.00 out of 1.00

Which tree traversal visits nodes in the order: left subtree, root, right subtree?

- a. In-order ✔
- b. Pre-order
- c. Post-order
- d. Level-order

Question 21

Correct

Mark 1.00 out of 1.00

What is the main function of the "push" operation in a stack?

- a. Adds an element to the top of the stack ✔
- b. Removes an element from the top of the stack
- c. Retrieves the bottom element
- d. Sorts the stack

Question 22

Correct

Mark 1.00 out of 1.00

What is the time complexity of inserting an element at the beginning of a linked list?

- a. O(1) ✔
- b. O(n)
- c. O(log n)
- d. O(n log n)

Question 23

Incorrect

Mark 0.00 out of 1.00

In the following scenarios, when will you use selection sort?

- a. Large values need to be sorted with small keys
- b. Small values need to be sorted with large keys
- c. The input is already sorted ✖
- d. A large file has to be sorted

Question 24

Correct

Mark 1.00 out of 1.00

What is the primary advantage of an adjacency list representation over an adjacency matrix?

- a. Uses less space for sparse graphs ✔
- b. Easier to implement
- c. Faster access time
- d. Uses more memory

Question 25

Correct

Mark 1.00 out of 1.00

What is the primary issue with hash tables when handling duplicate keys?

- a. Collisions occur more frequently ✔
- b. They take up more memory
- c. The hash function must be modified
- d. They require additional sorting operations

Question 26

Correct

Mark 1.00 out of 1.00

A tree structure is used to model an organizational chart where each manager can have at most two subordinates. What type of tree is this?

- a. Binary Tree ✔
- b. Ternary Tree
- c. Heap
- d. AVL Tree

Question 27

Correct

Mark 1.00 out of 1.00

Which data structure is best suited for implementing a priority queue?

- a. Heap ✔
- b. Stack
- c. Queue
- d. Linked List

Question 28

Correct

Mark 1.00 out of 1.00

In a singly linked list, what does each node contain?

- a. Data and a pointer to the next node ✔
- b. Data and a pointer to both previous and next nodes
- c. Only data without any pointer
- d. Multiple pointers to various nodes

Question 29

Correct

Mark 1.00 out of 1.00

Which of the following is a common hashing technique?

- a. Linear Probing ✔
- b. Quick Sort
- c. Binary Search
- d. Dijkstra's Algorithm

Question 30

Correct

Mark 1.00 out of 1.00

What is the time complexity of accessing an element at a known index in an array?

- a. O(1) ✔
- b. O(n)
- c. O(log n)
- d. $O(n^2)$

Question 31

Correct

Mark 1.00 out of 1.00

What does the Knuth-Morris-Pratt (KMP) algorithm use to avoid unnecessary comparisons?

- a. A prefix-suffix table (LPS (Longest Prefix Suffix) array) ✔
- b. A hash function
- c. Randomized searching
- d. A brute force approach

Question 32

Correct

Mark 1.00 out of 1.00

The process of removing a node with one child in a BST involves replacing it with ________.

- a. Its only child ✔
- b. The root node
- c. The maximum node
- d. The minimum node

Question 33

Correct

Mark 1.00 out of 1.00

What is the worst-case time complexity of BFS for a graph with V vertices and E edges?

- a. O(V + E) ✔
- b. O(V)
- c. O(E)
- d. $O(V^2)$

Question 34

Correct

Mark 1.00 out of 1.00

Which traversal method of a BST returns its elements in sorted order?

- a. In-order ✔
- b. Pre-order
- c. Post-order
- d. Level-order

Question 35

Correct

Mark 1.00 out of 1.00

Which of the following data structures is used to implement a priority queue?

- a. Heap ✔
- b. Stack
- c. Linked List
- d. Graph

Question 36

Correct

Mark 1.00 out of 1.00

In an unsorted array of size n, what is the worst-case time complexity of searching for an element?

- a. O(n) ✔
- b. O(log n)
- c. O(1)
- d. O(n log n)

Question 37

Correct

Mark 1.00 out of 1.00

Which traversal method visits nodes in sorted order when applied to a Binary Search Tree?

- a. In-order ✔
- b. Pre-order
- c. Post-order
- d. Level-order

Question 38

Correct

Mark 1.00 out of 1.00

What is the worst-case time complexity of the Knuth-Morris-Pratt (KMP) algorithm?

- a. O(n + m) ✔
- b. O(n^2)
- c. O(n log m)
- d. O(m log n)

Question 39

Correct

Mark 1.00 out of 1.00

Which sorting algorithm requires O(n) additional space due to its recursive implementation?

- a. Merge Sort ✔
- b. Quick Sort
- c. Selection Sort
- d. Bubble Sort

Question 40

Correct

Mark 1.00 out of 1.00

In a circular queue of size 5, if the rear is at position 4 and a new element is inserted, where will the rear move?

- a. 0 ✔
- b. 5
- c. 3
- d. 1